

# Bayesian regression for language scientists

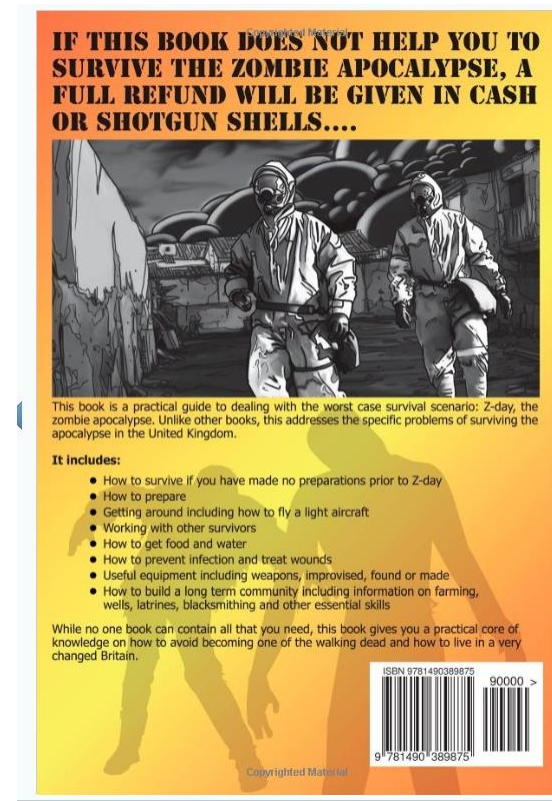
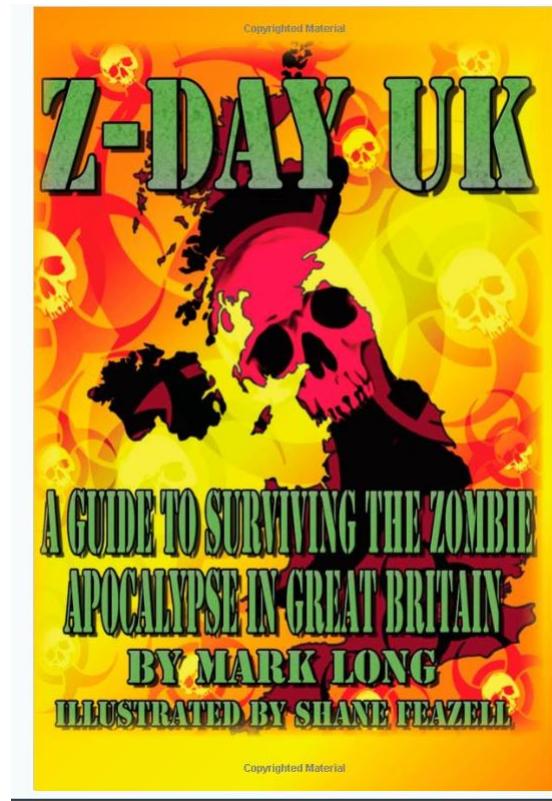
## Lecture 3

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# Case study: help + (to) Infinitive



*Here is how items in your emergency kit can **help** you **survive** the zombie apocalypse.*

<https://www.redcross.ca/blog/2013/2/preparedness-101-zombie-apocalypse>

## Preparedness 101: Zombie Apocalypse

Posted February 01, 2013 by [Jamie-Leigh Cuthbertson](#) - Photo aficionado

With the release of a zombie rom-com movie this weekend and AMC's The Walking Dead returning to our television screens shortly, we thought it might be a great time to share our zombie preparedness tips. Fuelled by a blog about zombies and emergency preparedness from the [US Centers for Disease Control and Prevention](#), North America has become almost obsessed with imagining how they would fare if the walking dead started roaming their neighbourhood.



Photo Credit: <http://www.cdc.gov/phpr/zombies>

Did you know that your [emergency kit](#) contains tools and supplies that can also be used in the event of an outbreak of the zombie virus? Here is how items in your emergency kit can help you survive the zombie apocalypse:

# Form of *help*

- The base form *help* occurs the most frequently with the bare infinitive.
- The form *helping* shows the highest proportion of *to*-infinitives.

*It is his job to see through the contracts that will **help rebuild** Iraq.*

*Many are partners with South African companies on projects that are **helping to rebuild** the country's infrastructure.*

# Distance between *help* and Infinitive

- The more words between help and the infinitive, the higher the chances of the *to*-infinitive:

*HUD will provide \$70 million for 1,300 rental vouchers to **help** people in public housing projects in Los Angeles, Chicago, Boston, Baltimore, and New York **to move** into surrounding middle-class and affluent suburbs.*

- This tendency can be explained by the principle of minimization of cognitive complexity: v“[i]n the case of more or less explicit grammatical options the more explicit one(s) will tend to be favoured in cognitively more complex environments” (Rohdenburg 1996: 151).

# *Horror aequi*

- A universal tendency to avoid repetition of identical elements.
- If the verb *help* is preceded by *to*, the second infinitive is usually without *to*:

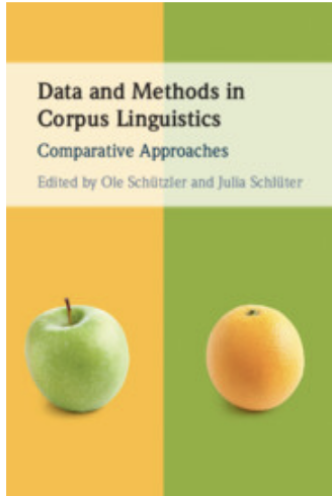
*With yoga, find a teacher who will make adjustments to **help prevent** injuries.*

# Helpee

- The chances of the marked infinitive are higher if the Helpee is implicit:

*[...] physical therapy has **helped** me **improve** my posture...*

*His encouragement and guidance **helped** [Ø] **to improve** my health and self-confidence.*



**Data and Methods in  
Corpus Linguistics**  
Comparative  
Approaches

## 8 - Comparing Bayesian and Frequentist Models of Language Variation

**The Case of Help + (to-)Infinitive**

**from Part III - Perspectives on Multifactorial Methods**

Published online by Cambridge University Press: **06 May 2022**

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**Chapter**



# Data: "Big" dataset

- 2050 examples from COCA
- Random effects: Verb
- Fixed effects:
  - Year\_new (transformed Year)
  - Horror: are there conditions for horror aequi effects?
  - Distance\_log: distance in words between help and the infinitive, log-transformed
  - Morph\_Form: morphological form of help
  - Helpee: the presence or absence of overt Helpee
  - Helper: whether the Helper is animate, inanimate or missing.

# Read the dataset in R

```
help_df <- read.table("help_big.txt", header = T,  
stringsAsFactors = TRUE)
```

```
str(help_df)
```

```
'data.frame': 2050 obs. of 10 variables:  
 $ Year      : int  2011 2002 1998 2009 2009 2001 2006 2006 2006 2003 ...  
 $ Response  : Factor w/ 2 levels "Bare","TO": 1 2 1 1 1 1 1 2 1 1 ...  
 $ Horror    : Factor w/ 2 levels "No","Yes": 2 2 1 1 2 2 2 1 1 1 ...  
 $ MorphForm : Factor w/ 4 levels "help","helped",...: 1 1 1 1 1 1 1 1 1 1 ...  
 $ Distance  : int   6 6 6 5 5 5 4 4 4 4 ...  
 $ Helpee    : Factor w/ 2 levels "No","Yes": 2 2 2 2 2 2 2 2 2 2 ...  
 $ Verb      : Factor w/ 594 levels "abate","absorb",...: 421 223 315 283 309  
38 454 351 195 361 ...  
 $ Helper    : Factor w/ 3 levels "Anim","Inanim",...: 3 1 2 2 3 3 1 1 2 2 ...  
 $ Year_new   : int   21 12 8 19 19 11 16 16 16 13 ...  
 $ Distance_log: num   1.79 1.79 1.79 1.61 1.61 ...
```

# Fitting a glmer model

```
library(lme4)
```

```
help_glmer <- glmer(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helppee + (1|Verb), data  
= help_df, family = binomial)
```

```
#convergence issues
```

```
help_glmer <- glmer(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helppee + (1|Verb), data  
= help_df, family = binomial,  
glmerControl(optimizer="bobyqa", optCtrl = list(maxfun =  
100000)))
```

## summary(help\_glmer)

...

Fixed effects:

|                            | Estimate | Std. Error | z value | Pr(> z ) |     |
|----------------------------|----------|------------|---------|----------|-----|
| (Intercept)                | -1.08956 | 0.17217    | -6.328  | 2.48e-10 | *** |
| Year_new                   | -0.05661 | 0.01046    | -5.415  | 6.14e-08 | *** |
| HorrorYes                  | -6.46193 | 1.56818    | -4.121  | 3.78e-05 | *** |
| Distance_log               | 0.85528  | 0.41933    | 2.040   | 0.04139  | *   |
| MorphFormhelped            | 0.50307  | 0.19266    | 2.611   | 0.00902  | **  |
| MorphFormhelping           | 3.14587  | 0.28112    | 11.190  | < 2e-16  | *** |
| MorphFormhelps             | 1.25889  | 0.24905    | 5.055   | 4.31e-07 | *** |
| HelpeeYes                  | -0.99159 | 0.39751    | -2.495  | 0.01261  | *   |
| HorrorYes:Distance_log     | 4.16950  | 1.33173    | 3.131   | 0.00174  | **  |
| MorphFormhelped:HelpeeYes  | 0.72105  | 0.37751    | 1.910   | 0.05613  | .   |
| MorphFormhelping:HelpeeYes | -1.80414 | 0.40270    | -4.480  | 7.46e-06 | *** |
| MorphFormhelps:HelpeeYes   | 0.23383  | 0.42164    | 0.555   | 0.57918  |     |

# C-index: measure of discriminative ability

- **0.5:** The model is no better than random chance (flipping a coin)
- **0.7 - 0.8:** acceptable
- **0.8 - 0.9:** good
- **0.9 - 1.0:** excellent

# Compute C-index

- We need predicted probabilities of TO-infinitive, for every observation.

```
glmer_predict <- predict(help_glmer, type = "response")  
head(glmer_predict)
```

| 1         | 2         | 3         | 4         | 5         | 6         |
|-----------|-----------|-----------|-----------|-----------|-----------|
| 0.3110800 | 0.4093780 | 0.2579603 | 0.1551730 | 0.1947764 | 0.2440183 |

```
length(glmer_predict)
```

```
[1] 2050
```

```
library(Hmisc)
```

```
somers2(glmer_predict, as.numeric(help_df$Response) - 1)
```

| C         | Dxy       | n            | Missing   |
|-----------|-----------|--------------|-----------|
| 0.8531138 | 0.7062275 | 2050.0000000 | 0.0000000 |

# Bayesian GLMM

```
library(brms)
```

A useful shortcut:

```
my_formula <- bf(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helpee + (1|Verb))
```



# The default priors

```
get_prior(formula = my_formula, data = help_df, family =
bernoulli)
```

| prior                | class     | coef                       | group | resp | dpar | nlpar | lb | ub | tag | source       |
|----------------------|-----------|----------------------------|-------|------|------|-------|----|----|-----|--------------|
| (flat)               | b         |                            |       |      |      |       |    |    |     | default      |
| (flat)               | b         | Distance_log               |       |      |      |       |    |    |     | (vectorized) |
| (flat)               | b         | HelpeeYes                  |       |      |      |       |    |    |     | (vectorized) |
| (flat)               | b         | HorrorYes                  |       |      |      |       |    |    |     | (vectorized) |
| (flat)               | b         | HorrorYes:Distance_log     |       |      |      |       |    |    |     | (vectorized) |
| (flat)               | b         | MorphFormhelped            |       |      |      |       |    |    |     | (vectorized) |
| (flat)               | b         | MorphFormhelped:HelpeeYes  |       |      |      |       |    |    |     | (vectorized) |
| (flat)               | b         | MorphFormhelping           |       |      |      |       |    |    |     | (vectorized) |
| (flat)               | b         | MorphFormhelping:HelpeeYes |       |      |      |       |    |    |     | (vectorized) |
| (flat)               | b         | MorphFormhelps             |       |      |      |       |    |    |     | (vectorized) |
| (flat)               | b         | MorphFormhelps:HelpeeYes   |       |      |      |       |    |    |     | (vectorized) |
| (flat)               | b         | Year_new                   |       |      |      |       |    |    |     | (vectorized) |
| student_t(3, 0, 2.5) | Intercept |                            |       |      |      |       |    |    |     | default      |
| student_t(3, 0, 2.5) | sd        |                            |       |      |      |       | 0  |    |     | default      |
| student_t(3, 0, 2.5) | sd        |                            | Verb  |      |      |       | 0  |    |     | (vectorized) |
| student_t(3, 0, 2.5) | sd        | Intercept                  | Verb  |      |      |       | 0  |    |     | (vectorized) |

# Fitting a GLMM with weakly informative priors

```
my_priors <- prior(cauchy(0, 2), class = b) +  
  prior(normal(0, 1), class = sd)
```

```
help_brm <- brm(formula = my_formula, data = help_df,  
family = bernoulli, cores = 4)
```

# Compare the betas

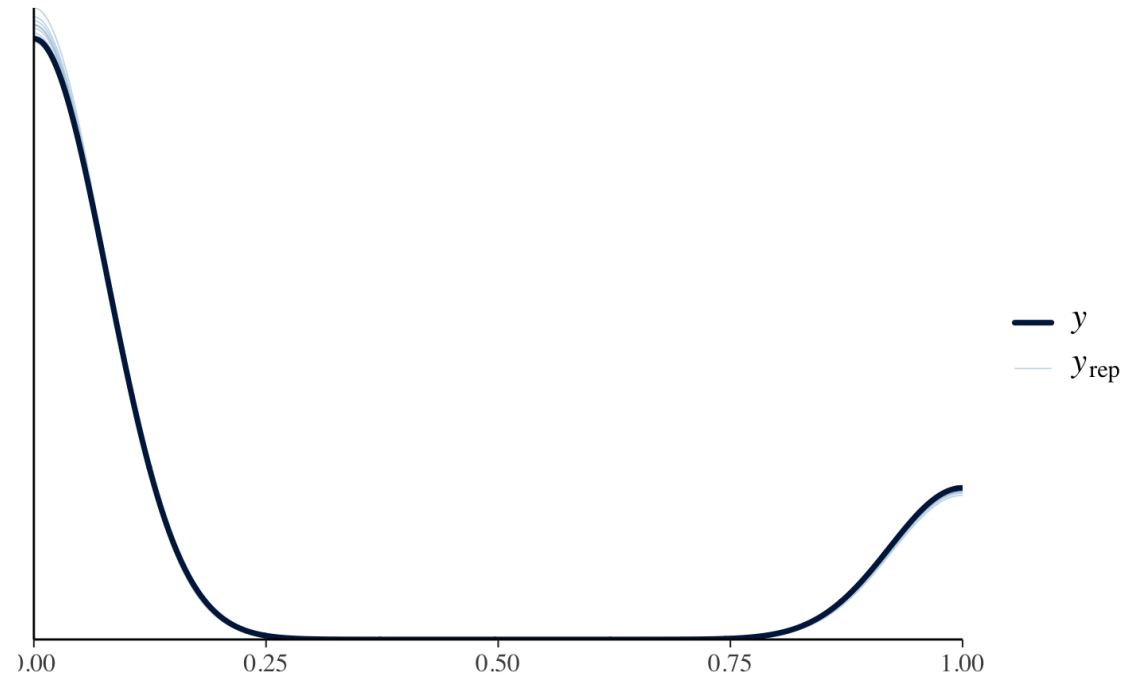
```
cbind(fixef(help_glmer) , fixef(help_brm) [, 1])
```

|                            | [,1]        | [,2]        |
|----------------------------|-------------|-------------|
| (Intercept)                | -1.08955621 | -1.10157816 |
| Year_new                   | -0.05661283 | -0.05777188 |
| HorrorYes                  | -6.46192823 | -7.01291861 |
| Distance_log               | 0.85527870  | 0.86764552  |
| MorphFormhelped            | 0.50306637  | 0.51314503  |
| MorphFormhelping           | 3.14587162  | 3.19713019  |
| MorphFormhelps             | 1.25888931  | 1.27838318  |
| HelpeeYes                  | -0.99159035 | -1.01295390 |
| HorrorYes:Distance_log     | 4.16950112  | 4.45869573  |
| MorphFormhelped:HelpeeYes  | 0.72105316  | 0.73925162  |
| MorphFormhelping:HelpeeYes | -1.80413945 | -1.83557619 |
| MorphFormhelps:HelpeeYes   | 0.23383437  | 0.22415471  |

# Posterior predictive check

- The traditional check for LM is probably not very meaningful for binary outcomes.

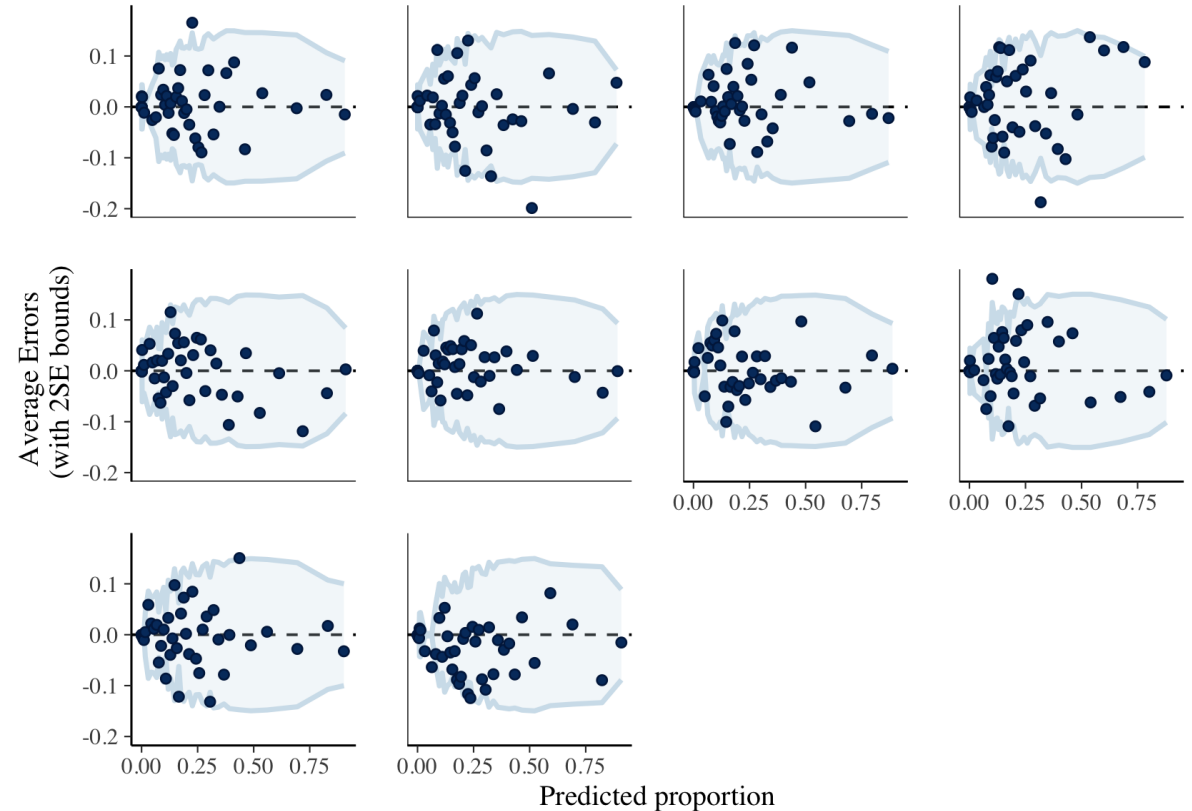
`pp_check(help_brm)`



# Posterior predictive check of binned errors

```
pp_check(help_brm, type = "error_binned")
```

- Points: bins of data points. Data points that the model assigned similar predictions to will be binned together.
- X-axis: predicted proportions of TO
- Y-axis: errors
- 95% of points should be within the blue regions



# Bayesian C-index

- To compute the C-index, we need the predicted probabilities of TO-infinite for every observation in the dataset (N = 2050)...
- AND for every posterior draw (N = 4000), because we are doing Bayesian statistics!

```
preds <- posterior_linpred(help_brm, newdata = help_df,  
transform = TRUE)
```

```
dim(preds)
```

```
[1] 4000 2050
```

# Probabilities of TO-infinitive

A fragment of the matrix:

`preds[1:5, 1:5]`

|       | [, 1]     | [, 2]     | [, 3]      | [, 4]      | [, 5]     |
|-------|-----------|-----------|------------|------------|-----------|
| [1, ] | 0.8179976 | 0.8494583 | 0.44975237 | 0.08342518 | 0.5000209 |
| [2, ] | 0.1778396 | 0.2292900 | 0.22343480 | 0.34528501 | 0.1421977 |
| [3, ] | 0.2730770 | 0.4466109 | 0.54543907 | 0.12741657 | 0.1790101 |
| [4, ] | 0.5849022 | 0.5252031 | 0.09837382 | 0.30683309 | 0.4291105 |
| [5, ] | 0.1462877 | 0.1402333 | 0.16004696 | 0.14338837 | 0.1557020 |

# Computing the Bayesian C-score

```
library(pROC)
for (i in 1:4000) {
  c_obj <- roc(response = help_df$Response, predictor =
preds[i, ], quiet = TRUE)
  c_scores[i] <- auc(c_obj)
}
mean(c_scores)
[1] 0.837815
quantile(c_scores, probs = c(0.025, 0.975))
2.5%      97.5%
0.8201381 0.8545324
```



# A practical challenge

- To incorporate a reviewer's suggestion, I decided to annotate the data for a very time-consuming variable, Stress (clash, lapse or nothing).
- Unfortunately, this was very time-consuming and difficult. I only managed to go through 400 examples.
- The new dataset may be too small...

# Horrors of horror aequi

```
help_glmer_small <- glmer(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helpee + Stress +  
(1|Verb), data = help_df_small, family = binomial,  
glmerControl(optimizer="bobyqa", optCtrl = list(maxfun =  
100000)))
```

```
summary(help_glmer_small)
```

...

|                        |           |            |        |          |
|------------------------|-----------|------------|--------|----------|
| HorrorYes:Distance_log | -21.65925 | 3460.01523 | -0.006 | 0.995005 |
|------------------------|-----------|------------|--------|----------|

# And other horrors...

```
optimizer (bobyqa) convergence code: 0 (OK)
```

```
unable to evaluate scaled gradient
```

```
Model failed to converge: degenerate Hessian with 2 negative  
eigenvalues
```

```
See ?lme4::convergence and ?lme4::troubleshooting.
```

```
Warning messages:
```

```
1: In vcov.merMod(object, use.hessian = use.hessian) :
```

```
variance-covariance matrix computed from finite-difference Hessian is  
not positive definite or contains NA values: falling back to var-cov  
estimated from RX
```

```
2: In vcov.merMod(object, correlation = correlation, sigm = sig) :
```

```
variance-covariance matrix computed from finite-difference Hessian is  
not positive definite or contains NA values: falling back to var-cov  
estimated from RX
```

# Data separation

```
table(help_df_small$Horror, help_df_small$Distance,  
help_df_small$Response)
```

```
, , = Bare
```

|     | 1   | 2  | 3 | 4 | 5 |
|-----|-----|----|---|---|---|
| No  | 136 | 71 | 7 | 1 | 0 |
| Yes | 52  | 45 | 2 | 1 | 0 |

```
, , = TO
```

|     | 1  | 2  | 3 | 4 | 5 |
|-----|----|----|---|---|---|
| No  | 64 | 13 | 4 | 1 | 2 |
| Yes | 1  | 0  | 0 | 0 | 0 |

# Bayesian models with different priors

- Non-informative, default priors for betas:

```
help_brm_flat <- brm(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helped + Stress +  
(1|Verb), data = help_df_small, family = bernoulli,  
cores = 4)
```

- Weakly informative priors for betas (see above):

```
help_brm_weak <- brm(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helped + Stress +  
(1|Verb), data = help_df_small, family = bernoulli,  
prior = my_priors, cores = 4)
```

# Bayesian model with strong priors

- The priors for the problematic interaction have the mean from the posterior means and standard deviation from Est. error

```
my_priors_strong <- prior(cauchy(0, 2), class = "b") +  
prior(normal(4.5, 1.5), class = "b", coef =  
"HorrorYes:Distance_log")
```

```
help_brm_strong <- brm(Response ~ Year_new +  
Horror*Distance_log + MorphForm*Helpee + Stress +  
(1|Verb), data = help_df_small, family = bernoulli,  
prior = my_priors_strong, cores = 4)
```

# Compare the coefficients in six models

```
cbind(big_glmer = fixef(help_glmer),  
      big_brms = fixef(help_brms)[, 1],  
      small_glmer = fixef(help_glmer_small)[-c(9:10)],  
      small_brms_flat = fixef(help_brms_flat)[-c(9:10), 1],  
      small_brms_weak = fixef(help_brms_weak)[-c(9:10), 1],  
      small_brms_strong = fixef(help_brms_strong)[-c(9:10), 1])
```

What is the effect of the strong prior?

Where do you still see the greatest discrepancies?

# Exercise

Try to add another strong prior based on the big dataset model. Will your choice of priors influence the posteriors?